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Project Report

Occlusion-Robust Face Identification Using a Pretrained ArcFace Model

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1. Introduction

Face-recognition systems perform well on clear, well-lit, frontal faces similar to their training images, but degrade sharply when identity-bearing regions are hidden by a mask, sunglasses, a scarf, a cap, or lighting/pose changes. This project addresses **closed-set face identification**: every test image belongs to an already-enrolled person, and the system returns the matching participant ID (e.g. p07) rather than just detecting an occlusion's presence.

The system uses a consent-based dataset of 30 participants captured under eight covering conditions and two lighting levels, processed through a fixed detect-align-crop pipeline, and classified with a **pretrained ArcFace (iResNet50) embedding** paired with a **RetinaFace** detector. Three experiments test how much task-specific training helps, and whether occlusion-diverse training data is necessary under compound occlusions. The system was packaged into a live phone-camera demonstration served from a laptop.

This report documents what was collected and built, and compares the approach against closely related published work.

2. Problem Statement and Objectives

2.1 Problem Statement

Given an image of a normal or partially occluded face, the system must classify it as one of 30 enrolled identities. Different occlusions hide different regions: a mask removes the nose/mouth/jawline, sunglasses the eyes, a cap the forehead/hairline, low light reduces texture generally. The central question: can a pretrained face-embedding model, given only a few hundred images per identity, stay discriminative under these conditions without being retrained on masks or sunglasses from scratch?

2.2 Objectives and Completion Status

1. Collect a structured face-image dataset covering a range of occlusion and lighting conditions. **Completed** — 30 participants, 6,000 raw images.
2. Capture each participant under eight covering conditions: none, cap, mask, sunglasses, scarf, clear glasses, mask+sunglasses, mask+clear glasses. **Completed.**
3. Detect, align, crop, resize, and quality-check every face image using one reproducible preprocessing pipeline. **Completed** — 5,988/6,000 images (99.8%) successfully detected and aligned.
4. Split the dataset into train/validation/test sets such that every identity, and where possible every identity-occlusion combination, appears in all three splits. **Completed.**
5. Evaluate a pretrained ArcFace pipeline with two forms of task-specific training: a trained classification head, and partial backbone fine-tuning. **Completed** — three experiments, E1–E3.
6. Evaluate overall accuracy, per-participant accuracy, and per-occlusion accuracy. **Completed** — extended beyond the original plan to also include per-occlusion precision/recall/F1 and a full 30×30 confusion matrix for every experiment (Section 9).
7. Produce a working demonstration that predicts the enrolled participant from a live phone camera. **Completed** — a browser-based demo streamed from a phone to a laptop over HTTPS on the local network.

3. Related / Base Papers

This project builds on three published works: one dataset/capture precedent, two recognition-method precedents. Table 1 summarises each; a detailed three-way comparison follows in Table 9.

Table 1: Base papers used for comparison.

Paper	Main contribution	Relation to this project
Salari and Rostami, “Pgu-Face: A Dataset of Partially Covered Facial Images,” <i>Data in Brief</i> , 2016 [3]	Introduces the Pgu-Face dataset: 896 images from 224 subjects, four images per subject, across up to two sessions at least six days apart, under clean-shaven/stubble conditions with and without sunglasses. Does not propose a recognition model.	Closest public dataset precedent for identity-labelled, partially-covered face capture with a controlled protocol; this project’s dataset (6,000 images, 30 identities, 8 occlusion types) is far larger in image count and occlusion variety, though over far fewer identities and (unlike Pgu-Face) a single session.
Chong, Chong, and Ong, “Masked Face Recognition Using Support Vector Machine and Convolutional Neural Network,” <i>ICoICT</i> , 2022 [4]	Trains a CNN for feature extraction and an SVM for label classification on masked-face benchmarks (RMFD, LFW-SMFD); reports 98.39% TAR on a 50-category RMFD subset, with a marked accuracy drop as the category count grows, since SVM classification does not scale well to more identities.	Closest published recognition-method precedent using a trained classifier on top of extracted features; this project instead adapts a pretrained ArcFace embedding rather than training a CNN feature extractor from scratch, and trains a lightweight identity head instead of an SVM.
Montero, Nieto, Leskovsky, and Aginako, “Boosting Masked Face Recognition with Multi-Task ArcFace,” <i>SITIS</i> , 2022 [5]	Extends an ArcFace-based ResNet50 pipeline with masked-face training augmentation and an auxiliary mask-detection output, so the same network both recognises identity and predicts whether a mask is present.	Directly comparable ArcFace-family model adapted specifically for masks at larger scale; shows what mask-specific architecture changes buy over the untouched pretrained embedding this project starts from, and is the closest precedent for treating occlusion-diverse training data as the key lever, matching this project’s own E1-vs-E2 finding (Section 9).

4. Dataset

4.1 Team and Participants

Thirty identities (p01–p30) were enrolled: six team members each recruited and photographed five people (themselves plus four consenting relatives/acquaintances). The participant-to-photographer mapping lives in `metadata/person_id_mapping.txt`, kept internal rather than reproduced here.

4.2 Consent

Every participant signed a consent form before any photograph. It explains this is an academic, non-commercial prototype; each participant gets an anonymous ID (e.g. p07), never a name; raw images and the identity mapping stay restricted to the six members; participation is voluntary and withdrawable; and a recognisable photo appears in the report/demo only for participants who separately tick an optional permission box.

4.3 Capture Protocol

Each participant was photographed under eight covering conditions — **none**, **cap**, **mask**, **scarf**, **clearglass**, **sunglass**, **mask_clearglass**, **mask_sunglass** — crossed with two lighting levels, **regular** and **low**, in a single sitting, with head position varying naturally across shots rather than a logged pose label. The dataset totals **6,000 images**: 30 identities $\times$ 200 images, well beyond the proposal’s original 10–15-participant target.

Filenames follow a fixed pattern (e.g. p07_mask_sunglass_low_014.jpg), so identity, occlusion, and lighting parse without a lookup table; a manifest CSV is the authoritative metadata source.

4.4 Pose, Lighting, and Obstruction Categories

Table 2 records the informal head-pose variation during capture (not logged per-image); Tables 3 and 4 record the two dimensions logged for every image, since occlusion and lighting are what the task is actually evaluated against.

Table 2: Head-pose variation encouraged during capture.

Label	Meaning
front	Looking directly at the camera.
left / right	Head turned to the participant’s left / right.
up / down	Chin tilted upward / downward.

Table 3: Lighting labels used during capture.

Label	Meaning
regular	Even indoor lighting; the face is clearly visible.
low	Dim lighting; the face is dimmer but still detectable.

Table 4: Obstruction (covering) labels used during capture.

Type	Labels
Single covering	none , mask , sunglass , clearglass , scarf , cap
Combined covering	mask_sunglass , mask_clearglass

Figure 1 shows one accepted example per occlusion family; Figure 2 gives the per-condition image count after preprocessing – close to, but not perfectly, balanced, since a few images per condition were lost to quality control (Section 5).



Figure 1: Real accepted samples, one per occlusion family (participants who ticked the optional photo-permission box).

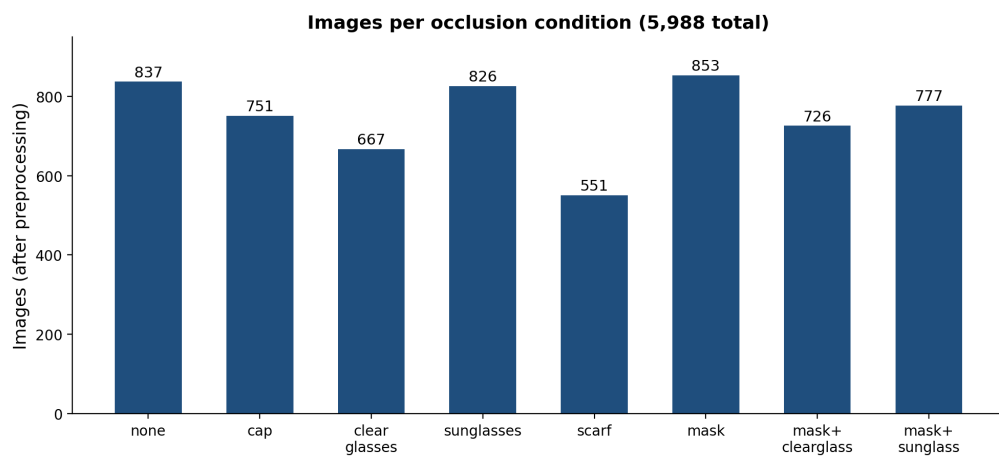


Figure 2: Number of images per occlusion condition after preprocessing (5,988 total).

4.5 A Note on Synthetic Images

The large majority of the dataset is real photographs: every identity was photographed wearing every occlusion in person. A small minority are an exception: a few participants declined to physically wear a specific occlusion (most often sunglasses, for personal or religious reasons), so that occlusion was digitally added to one of their clean photos instead, just so that identity's coverage would not be entirely missing. These are a small, unflagged fraction of the dataset – disclosed here, not claimed to be common.

Separately, the training-only augmentation pipeline (Section 8) applies mild synthetic transformations — flips, rotations, brightness/contrast jitter, resolution degradation, blur, sensor noise, JPEG re-compression, random erasing — to simulate the live phone-camera domain. These are training-time variations of real photos, not new identities or occlusions.

5. Data Cleaning and Quality Control

Rejected photos move to `put_asides/` with a recorded reason, never deleted: blur, low light, full coverage, or bad framing. Figure 3 shows the flow; Figure 4 shows one real example per reason. The dataset is frozen from the start of training, so every number here comes from the same 6,000-image set.

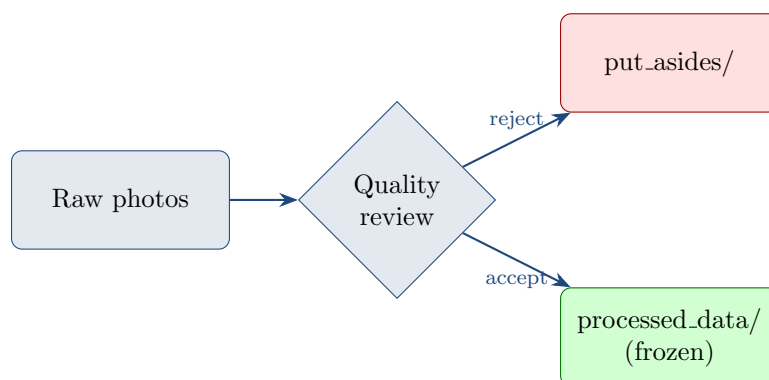


Figure 3: Data cleaning and quality-control flow: every captured photo is either rejected with a recorded reason or passed through to the frozen dataset.



Figure 4: Real rejected samples, one per quality-control reason.

6. Data Preprocessing

Every accepted photo passes through one fixed pipeline (`scripts/01_preprocess.py`), identical for training, validation, test, and later, the live demo. Figure 5 shows the flow.

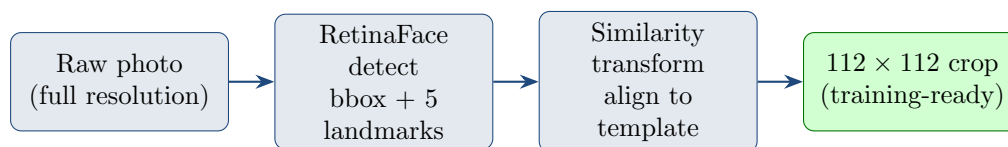


Figure 5: Preprocessing pipeline: every accepted photo, before it can be used for training, evaluation, or live inference.

1. **Detect.** RetinaFace (`insightface`'s `buffalo_l` detector, `det_10g.onnx`) locates the face and its five landmarks at `det_size=(640,640)`, `det_thresh=0.5`. The largest detected face is kept.
2. **Fallback on failure.** If nothing is detected, the image is retried at `det_thresh=0.3`, then at $2\times$ upscale. Every failure is logged, never dropped silently.
3. **Align and crop.** A similarity transform (`insightface.utils.face_align.norm_crop`) maps the five landmarks onto ArcFace's reference template, producing a 112×112 aligned crop – this is what stops the classifier keying on background or camera artifacts.
4. **Log.** Each image's path, identity, occlusion, lighting, MD5 hash, and detection outcome are appended to `metadata/preprocess_log.csv`.

Result: 5,988 of 6,000 images (99.8%) were successfully detected and aligned, above the 97% acceptance gate. The 12 failures concentrated in the hardest conditions — 10 in `mask_sunglass`, 2 in `mask` — where too little face remained visible for landmark detection even at the loosened threshold.

7. Data Organisation and Split

7.1 Group-Aware Splitting

Splitting is not simple random assignment: near-duplicate images (a burst of photos seconds apart, or a low-light image derived by darkening a regular one) could otherwise land on opposite sides of the split and inflate test accuracy. (Exact, byte-identical duplicates from repeat shutter presses also exist and are treated as independent images; every MD5 hash is logged for auditability.) Two images are grouped, and forced into the same split, if:

- they are derived from the same source image, or
- they share a person *and* occlusion and were captured within a 10-second window, with burst-chains capped at 8 seconds to prevent one long burst from swallowing an entire condition.

Groups are then assigned as whole units via a deficit-based greedy allocator: largest groups first, each into whichever split has the biggest shortfall against its 60/20/20 target, per (person, occlusion) cell. This avoids the naive sequential-fill failure seen during development, which produced a wildly unbalanced split (train 24%, val 58%, test 18%) from a single 49-image group.

7.2 Coverage Guarantee

A hard requirement was enforced: **every participant must appear in all three splits**. A repair pass can force-move whole groups from train into an under-represented split if needed; it was not triggered in the final run. Per-occlusion coverage per split is best-effort only: 36 of 240 (person $\times$ occlusion) cells are absent from test specifically (informational, not a failure).

7.3 Final Split

Table 5: Final train/validation/test split (5,988 successfully preprocessed images).

Split	Images	Share
Train	3,546	59%
Validation	1,444	24%
Test	998	17%
Total	5,988	100%

All hard assertions (no group spans two splits; every participant in all three splits) passed on this final assignment.

8. Model and Training Approach

8.1 Selected Models

- **RetinaFace** [2] (`det_10g.onnx`) — detection and five-point landmark alignment (Section 6).
- **Pretrained ArcFace** [1], iResNet50 backbone (`w600k_r50.onnx`, `insightface buffalo_l`) — turns each aligned 112×112 crop into a 512-dimensional embedding. This is the primary identification model in every experiment below.

RetinaFace and ArcFace were selected because they are purpose-built for faces, need far less local data than training from scratch, run on both the remote GPU box and a laptop CPU, and have precedent adapting the same family to occluded faces (Section 3). No from-scratch CNN baseline was trained: with only 30 classes and a few hundred images per identity, it was judged inherently weaker than adapting a pretrained backbone – effort went to the fine-tuning ladder below instead.

8.2 Experiments

Three experiments were run, each answering a specific question about how much task-specific training the pretrained embedding actually needs:

E1 and E2 keep the pretrained ArcFace backbone frozen and train only a new classification layer on top, using the same family of loss ArcFace itself was trained with. The two runs are identical except their training data: E1 sees only clean, unoccluded faces; E2 sees all eight occlusion conditions. Figure 6 shows the pipeline.

E3 goes further, unfreezing the last 15% of the backbone’s own parameters alongside a new identity head, letting the network slightly adjust its face representation instead of only fitting a boundary on a fixed one. Only the last layers are unfrozen: with just a few thousand training images, adjusting the whole backbone risks overwriting its general face knowledge, and the final layers encode the most identity-specific detail anyway. Batch-normalisation statistics stay

Table 6: Experiment design.

ID	Name	Backbone	Question
E1	Trained head, clean-only	Frozen + new identity head	The proposal’s original baseline: train only on the none condition.
E2	Trained head, mixed-condition	Frozen + new identity head	Does training on all eight occlusion conditions help over E1?
E3	Partial backbone fine-tune	Last 15% of backbone unfrozen + head	Does adapting backbone weights (not just the head) help further?

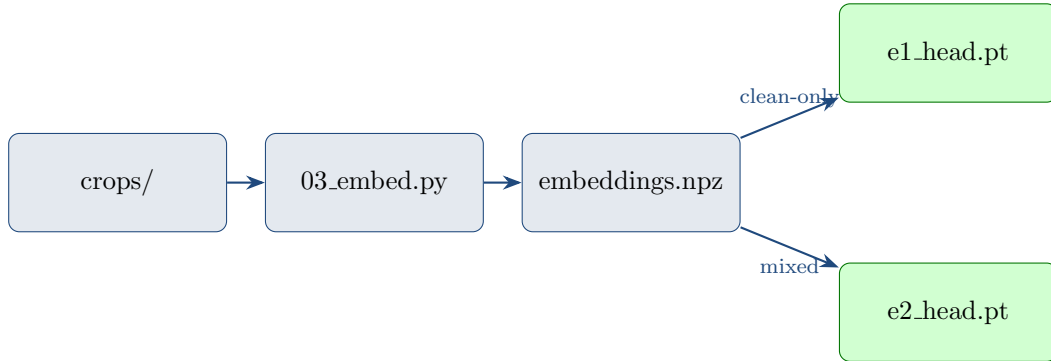


Figure 6: E1/E2 training pipeline: a shared frozen embedding, two heads trained on different data.

frozen throughout, since updating them on so little data was found to degrade the pretrained features. Since *insightface* ships ArcFace as inference-only, the backbone was converted to a trainable form and verified to match the original (outputs agreed to four decimal places) before training. Figure 7 shows the pipeline.

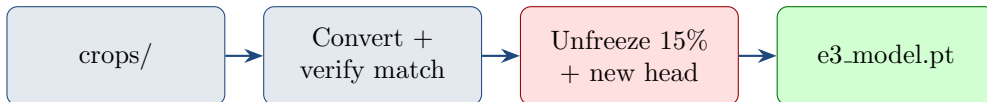


Figure 7: E3 training pipeline. Raw crops are needed again here, since gradients must flow back through the backbone.

How the validation split was used. It never contributes to any reported accuracy, but does two jobs: deciding when to stop training (once validation performance stops improving, so the model isn’t over-memorising train data), and guiding setting choices during development — margin strength, learning rates, how much backbone to unfreeze — by comparing candidates on validation before ever touching test data.

No augmentation was applied to validation or test data at any stage; the live-capture-style augmentations described in Section 4.5 were applied only to E1/E2/E3 training images.

9. Results

Because every identity contributes an equal, fixed share of the dataset (Section 4), plain top-1 accuracy is already a fair headline number overall. Within a single occlusion, though, one accuracy figure can hide *which* identities drive the errors, so the per-occlusion breakdown below reports precision/recall/F1 instead: computed only from that occlusion’s test images, so preci-

sion specifically penalises identities over-predicted within that condition. Each experiment was also evaluated with `scripts/10_full_eval.py`, which computes a full 30×30 confusion matrix, kept under `runs/eval/<experiment>/`.

Table 7: Test-set top-1 accuracy across all three experiments (998 test images, 30 balanced classes).

Experiment	Top-1 accuracy
E1 — Head, clean-only	94.29%
E2 — Head, mixed-condition	99.90%
E3 — Partial backbone fine-tune	100.00%

Table 8: Per-occlusion precision (P), recall (R), and F1, across all three experiments, showing exactly where each approach’s remaining errors are concentrated. Each triple is computed only from that occlusion’s test images.

Occlusion	n	E1			E2			E3		
		P	R	F1	P	R	F1	P	R	F1
none	138	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
cap	129	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
clear glasses	108	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
sunglasses	125	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
scarf	84	97.4	96.3	96.8	100.0	100.0	100.0	100.0	100.0	100.0
mask	161	97.3	96.0	95.8	99.5	99.4	99.4	100.0	100.0	100.0
mask+clear glasses	120	87.6	86.4	85.8	100.0	100.0	100.0	100.0	100.0	100.0
mask+sunglasses	133	75.5	71.4	68.9	100.0	100.0	100.0	100.0	100.0	100.0

E1 vs. E2 is the clearest, most informative comparison here. E1 (clean-only) drops to 75.5% precision / 71.4% recall / 68.9% F1 specifically on **mask+sunglasses** — the one pair hiding both the eyes and the lower face — while scoring a perfect 100.0% on every single-occlusion condition. E2 (all eight conditions) recovers to a perfect 100.0% on that same condition and 99.90% top-1 overall. This is direct evidence that occlusion-diverse training data matters specifically for compound occlusions, not single ones, which a pretrained embedding already handles well unaided.

E3 reached 100.00% top-1, correctly classifying every test image. Since E2 was already at 99.90%, E3’s gain is small – it closes the last gap rather than solving one E2 left substantially open. With both at or above 99.9%, the more informative result here is the E1-vs-E2 comparison above, not the marginal E2-vs-E3 gap.

9.1 Confusion Matrices

Figures 8 and 9 show the row-normalised confusion matrix for E1 and E2 (E3’s is a perfect diagonal and is omitted as uninformative). E2 is a clean diagonal with one isolated cell; E1 is the only one with a systematic error pattern, concentrated under **mask+sunglasses** – most often **p06→p16** (6 images), **p16→p11** (5), **p27→p11** (4), all resolved once E2’s occlusion-diverse training was used. Full counts for every experiment are kept in `runs/eval/<experiment>/confusion_matrix.csv`.

10. Comparison with Related Work

Table 9 places this project side by side with its two base papers (Section 3): Pgu-Face, the closest dataset precedent, and Chong et al., the closest method precedent. The comparison is necessarily approximate, since neither evaluates the same task, scale, or identity pool.

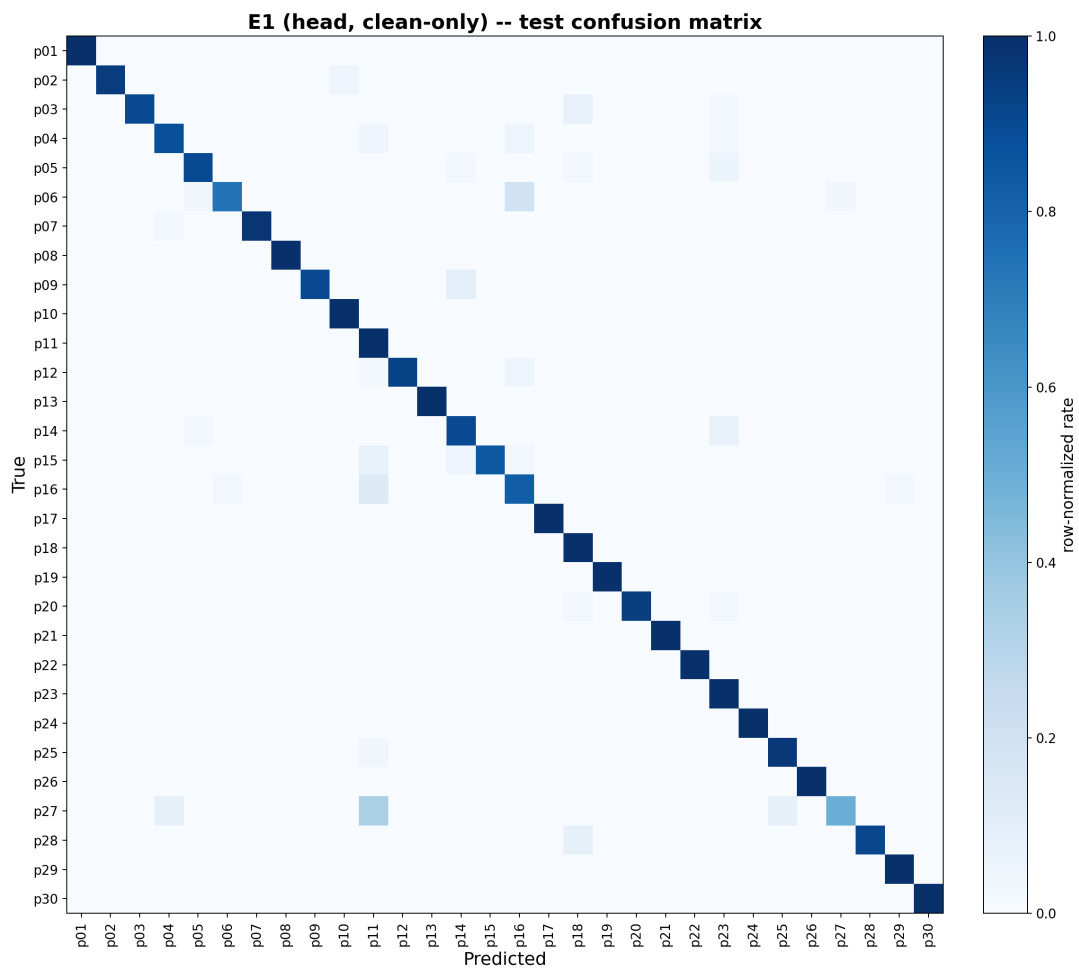


Figure 8: Row-normalised 30×30 test confusion matrix, E1 (head, clean-only). Darker cells indicate a higher share of that identity's test images landing in that predicted column; a perfect classifier is a solid diagonal.

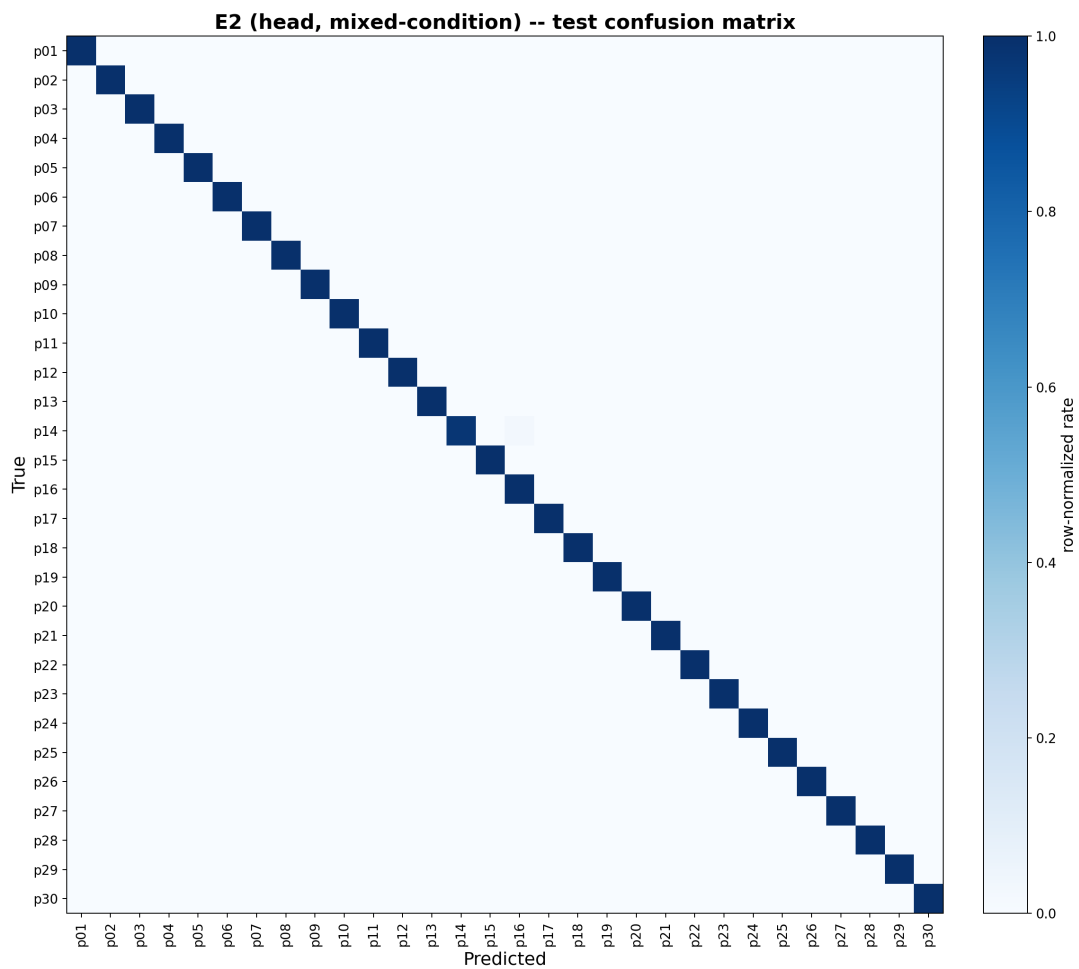


Figure 9: Row-normalised 30×30 test confusion matrix, E2 (head, mixed-condition).

Table 9: Detailed three-way comparison: Pgu-Face (dataset base), Chong et al. (method base), and this project.

Aspect	Pgu-Face [3]	Chong et al. [4]	This project
Paper type	Dataset / Data in Brief article	Recognition method paper	Applied ML system
Primary objective	Create a partially-covered face dataset	Recognise masked faces using CNN + SVM	Identify known people under multiple real occlusions
Identities	224	RMFD 525 / LFW-SMFD 5,713 full; main experiments use 50	30
Images	896	90,000 (RMFD); 13,027 (LFW-SMFD); subsets used for experiments	6,000 captured / 5,988 processed
Occlusion conditions	Sunglasses + facial-hair conditions	Primarily masks	Masks, sunglasses, clear glasses, scarves, caps, combinations
Capture source	Several consumer/mobile and commercial cameras	Public benchmark datasets	Local real capture (phone cameras)
Face detection	Not a proposed pipeline	Not central to the method	RetinaFace
Alignment	Not proposed as a recognition pipeline	Not central	Five-point landmark alignment
Feature representation	No proposed recognition representation	CNN feature vector	512-D pretrained ArcFace embedding
Recognition model	None proposed	CNN + SVM	Pretrained ArcFace + 30-way local head
Training model	No recognition training model	CNN trained for feature extraction; SVM classifies labels	Head training; optional partial backbone fine-tuning
Training data strategy	Dataset construction	Masked + unmasked customised training	Clean-only vs. mixed-condition vs. partial fine-tuning
Epochs	Not applicable	100, in reported experiments	Early stopping for E1/E2; fine-tuning for E3
Best reported recognition result	Not applicable	98.39% TAR (RMFD, 50-category); 95.24% TAR (LFW-SMFD, 50-category)	99.90% (E2); 100.00% (E3)
Scalability	Dataset-oriented	Strong degradation as SVM category count grows	Not tested at large-scale / open-set
Deployment	None	Research evaluation only	Live phone-browser demo over HTTPS

Aspect	Pgu-Face [3]	Chong et al. [4]	This project
Main value to this project	Dataset/capture precedent	Algorithmic/method precedent	Modern pretrained adaptation + broader occlusion + live deployment

This project does not “beat” either base paper — Pgu-Face proposes no recognition model, and Chong et al.’s 98.39% TAR comes from a different dataset, protocol, and a 50-identity SVM setting that the same paper shows degrading as identities grow. Its own strongest controlled comparison is E1 (94.29%) to E2 (99.90%) — same architecture and data scale, occlusion-diversity the only change — a fair result no cross-paper comparison can offer. The contribution is a practical adaptation: a modern pretrained representation over a from-scratch CNN+SVM pipeline, a broader real occlusion set, and a working live deployment neither base paper attempts.

11. Deployment — Live Demonstration

11.1 Model Selection and Live Switching

A dropdown selects between E1, E2, and E3 live, without restarting the server. **E2 is the default**, since it is close to E3’s accuracy (99.90% vs. 100.00%) without needing E3’s heavier backbone loaded first. A per-identity *gallery* classifier with a calibrated UNKNOWN-rejection threshold (0.3179) is implemented (`scripts/07_build_gallery.py`) but currently disabled in the dropdown; consequently E1/E2/E3 are all closed-set only – every detected face is assigned one of the 30 identities, with no mechanism yet active to flag an unenrolled visitor.

11.2 Live Phone-Camera Demo

The demo runs as a Flask server on a laptop, serving a browser page to a phone over the local Wi-Fi network via HTTPS. The phone’s camera streams frames roughly every 400 ms (900 ms for E3); each frame is detected, aligned, embedded by whichever model is selected, averaged over a rolling buffer (cleared on mode switch, since E3 uses a different embedding space), and classified. The result renders on the phone screen in real time. A camera-lens picker and front/back toggle are included, since many phones default to a blurry ultra-wide rear lens. Figure 10 shows the idle state; Figure 11 shows the full pipeline.

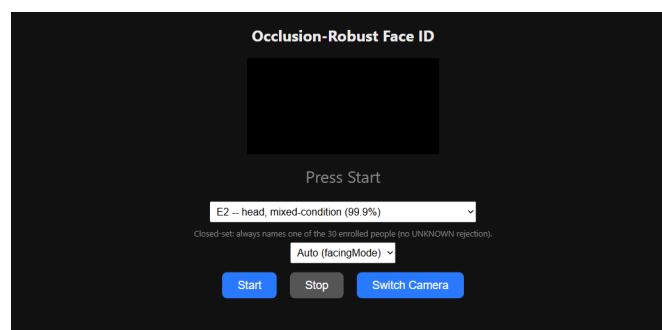


Figure 10: The live demo’s web interface (idle state, before the camera is started), captured directly from the running server.

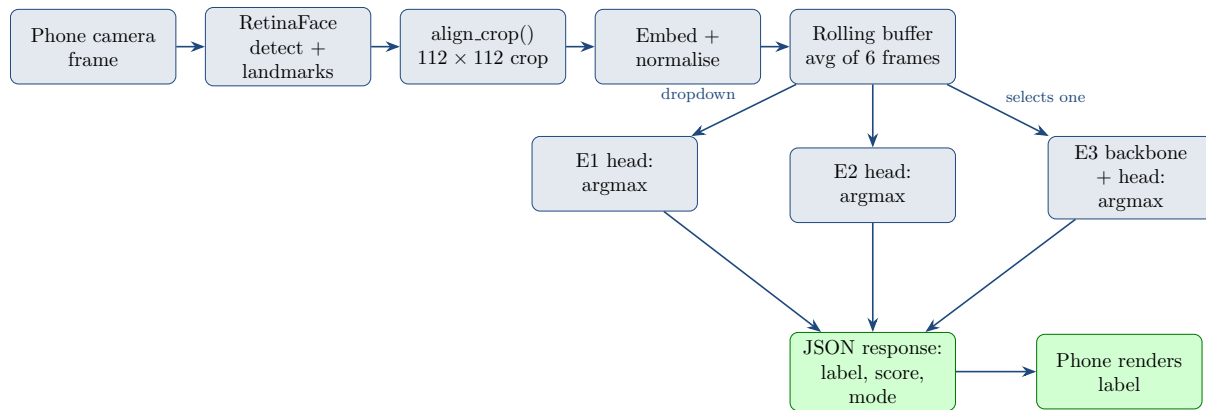


Figure 11: Live inference pipeline: what happens between one camera frame arriving at the server and a label being sent back to the phone. The rolling buffer’s averaged embedding is classified by whichever of E1/E2/E3 is currently selected in the dropdown (Section 11); the other two branches are not evaluated for that frame.

12. Limitations and Future Work

- **Closed set only.** Every input is forced into one of 30 identities (E1/E2/E3, none reject an unenrolled face); a gallery-based classifier with a calibrated rejection threshold exists but is disabled (Section 11). No formal open-set evaluation was performed.
- **30 identities.** Results will not extrapolate to hundreds or thousands of enrolled people, and are not comparable in absolute terms to the open-set results of Section 10.
- **No anti-spoofing or liveness detection.** A laboratory prototype, not an access-control system.
- **Future work** suggested by Section 3’s base papers: an auxiliary mask-detection objective alongside identity [5], or a larger identity count to test whether [4]’s SVM-style scaling limits have any analogue here – both testable on a second, multi-session capture round.

13. Supplementary Materials

The project repository includes three companion documents kept in sync with the code rather than restated here:

- `docs/arch.md` — four diagrams: manual data curation, the constrained split, the E1–E3 training pipeline, and the live-demo inference pipeline, labelled with actual repository file/folder names.
- `docs/inference.md` — a step-by-step walkthrough from camera frame to returned label, with one worked example carried through every stage.
- `docs/models.md` — a one-table reference of every model used, where, and why.

A 20-slide presentation (`docs/presentation.pptx`) is also included, covering the same material with native, editable diagrams and the results charts from Section 9.

14. Conclusion

This project built and evaluated a closed-set face-identification system robust to real occlusions, using a pretrained ArcFace model. Thirty participants were recruited, and 6,000 images collected across eight occlusion conditions and two lighting levels — well beyond Pgu-Face’s scale. A reproducible pipeline processed 99.8% of the dataset, and a duplicate-aware, coverage-guaranteed split produced a 3,546/1,444/998 partition. Three experiments showed that a head trained only on clean faces degrades specifically on **mask+sunglasses** (E1, 94.29% overall, 71.4% on that condition), training on occlusion-diverse data recovers it (E2, 99.90%), and partial backbone fine-tuning closes the remaining gap entirely (E3, 100.00%). The system was deployed as a live phone-camera demonstration, defaulting to E2 with E1/E3 selectable via dropdown. Against its two base papers, this project is not offered as beating either — Pgu-Face proposes no recognition model, and Chong et al.’s result comes from a different dataset and protocol — but as a practical adaptation: a pretrained representation over a from-scratch CNN+SVM pipeline, a broader occlusion set, and a working live deployment neither attempts. The honest contribution is the controlled E1-vs-E2 result: occlusion-diverse training data, not architecture or loss engineering, closed the gap on the hardest occlusion condition.

References

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